ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** P	The pin falls first.
2	**(d)** B	Translucent materials.
3	**(d)** its shape displaces much water	Hollow shape lowers average density.
4	**(d)** S	The pin falls last.
5	**(a)** C conducts heat and D does not	Test for conduction.
6	**(b)** it is light and cushions shocks	Soft and light.
7	**(a)** iron is heavier for the same size	Comparing same volume.
8	**(c)** using a magnet	Magnet only affects iron; keeps others intact.
9	**(a)** Both A and R are true and R is the correct explanation of A.	We can see clearly through glass because light passes through it; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are about different materials; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Empty bottle has low average density (1); Sand makes it heavier (1).	1+1
12	Rubber is an electrical insulator (1); Prevents electric shock (1).	1+1
13	Shaping changes the object (1); E.g. steel gives spoon, pan or bridge parts (1).	1+1
14	Metal conducts heat (1); Plastic is an insulator (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It is a metal (1); Shiny (lustrous) and sinks (1); Conducts heat along its length (1).	1+1+1
16	Prevent shock (1); Rubber or plastic (1); Insulate the hand (1).	1+1+1
17	Metals are strong (1); Metals give a ringing sound when struck (1); Metals can be shaped (1).	1+1+1
	OR: Wrap one hot container in cotton and other bare (1); Measure temperature after 15 min (1); Cotton traps air and keeps hot longer (1).	1+1+1
18	Light (1); Cheap (1); Waterproof and can be moulded (1).	1+1+1
	OR: First (1); It absorbs more (1); Larger absorbency (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Steel (a metal) (1). (ii) Glass (1). (iii) Wood (1). (iv) Metal conducts heat well, so it heats the food (1); wood is a poor conductor so the handle stays cool to hold (1).	1+1+1+2
20	Magnet for iron (1); Add water to dissolve salt and dye (1); Filter for sand (1); Evaporate to get salt and dye mix (1); Principle: magnetism, solubility, evaporation (1).	1+1+1+1+1
	OR: Cloth: waterproof (1); Metal rods: strong (1); Plastic handle: insulator (1); Light weight for carrying (1); Foldable parts (1).	1+1+1+1+1

Mistakes to avoid:

- Calling frosted glass or tracing paper transparent; they are **translucent**.
- Saying that insoluble means the substance cannot mix with anything; it means it does not dissolve in **water**.
- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.
- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** sand on the filter paper	Sand is insoluble.
2	**(c)** heating suddenly	Sudden heat can crack it.
3	**(c)** a material that is opaque may be shiny	Metals are opaque and shiny.
4	**(b)** evaporate the water	Evaporation leaves salt.
5	**(b)** it is light and catches wind	Light fabric.
6	**(c)** temperature affects dissolving	Hot water dissolves faster.
7	**(d)** the properties of the materials	Choose by properties.
8	**(c)** strength and durability	Strong materials carry loads.
9	**(d)** A is false but R is true.	A is false: sand is insoluble. R is true.
10	**(c)** A is true but R is false.	A is true; R is false because most metals are hard (sodium is an exception).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Uneven expansion cracks glass (1); Glass is a poor conductor so heats unevenly (1).	1+1
12	Wool traps air and keeps heat (1); It absorbs water (1).	1+1
13	Add each to water (1); One dissolves, one stays (1).	1+1
14	They can be beaten into thin sheets (1); They can be bent and shaped (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnet removes iron (1); Water: sawdust floats, skim off (1); Filter sand, evaporate salt water (1).	1+1+1
16	Same temperature (1); Same stirring (1); Time taken to dissolve (1).	1+1+1
17	Heat gently (1); Wax melts (1); Pour off or filter hot (1).	1+1+1
	OR: Wood is a poor conductor (1); Steel conducts quickly (1); Used for stirring safely (1).	1+1+1
18	Allows part of light (1); Image seen blurred (1); Frosted glass (1).	1+1+1
	OR: Metal body conducts heat (1); Strong to hold pressure (1); Plastic handles insulate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Salt and sugar (1). (ii) Sand (or chalk) (1). (iii) Oil (1). (iv) Oil does not mix with water and is lighter than it (1); salt (or sugar) dissolves completely (1).	1+1+1+2
20	Spoon, pan: steel (1); Window, bottle: glass (1); Chair, door: wood (1); Pen, toy: plastic (1); Same material many forms (1).	1+1+1+1+1
	OR: Choose five materials (1); Test lustre (1); Test hardness (1); Test solubility (1); Table and conclusion (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.
- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** it is strong and cheap	Natural fibre.
2	**(a)** raincoats	Waterproof.
3	**(a)** it can be rolled into thin sheets and is non-toxic	Malleable.
4	**(d)** a brick	Brick.
5	**(d)** innerwear	Absorbent.
6	**(b)** lustre	Both reflect.
7	**(d)** poor heat conductor and non-conductor of electricity	Safe to touch.
8	**(c)** using glass for a greenhouse	Glass lets sunlight in.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Because metals conduct heat well, the spoon takes heat from the tea and passes it along.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Poor conduction keeps the handle from becoming too hot, so R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is less dense than water (1); It floats (1).	1+1
12	Magnetic property for iron (1); Solubility for salt (1).	1+1
13	Sugar dissolves in water (1); Pot would not hold water (1).	1+1
14	Paper absorbs water and becomes weak (1); Water leaks through the soggy paper (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Check absorbency with water (1); Cotton absorbs water (1); Nylon does not (1).	1+1+1
16	For a fair test (1); Different conditions change results (1); Valid comparison (1).	1+1+1
17	They absorb or scatter some light (1); Edges bend light (1); Coloured glass casts a coloured shadow (1).	1+1+1
	OR: Iron is strong and hard (1); Cotton is soft and absorbs sweat (1); The use depends on the property (1).	1+1+1
18	Glass and some plastics also shine (1); Metals also conduct heat (1); So several properties are needed (1).	1+1+1
	OR: Paint protects from rusting (1); Gives colour (1); Makes surface smooth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Lustre (1). (ii) Group A (1). (iii) Wood (1). (iv) Copper is a good conductor of electricity so current flows easily (1); plastic is an insulator, so it protects us from shock (1).	1+1+1+2
20	Gold is soft (1); Diamond is hardest (1); Gold can be scratched with a hard object (1); Gold is shiny and malleable (1); Claim is false (1).	1+1+1+1+1
	OR: Saves natural resources (1); Reduces waste (1); Saves energy (1); Reduces pollution (1); Gives jobs (1).	1+1+1+1+1

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